

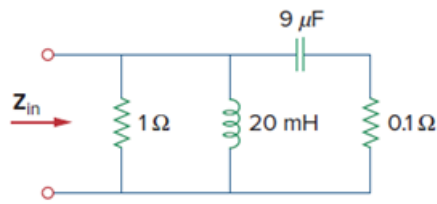
Problem

For the circuit in Fig. 14.83, find:

(a) the resonant frequency ω_0

(b) $Z_{in}(\omega_0)$

Figure 14.83:



Step-by-step solution

Step 1 of 5

Refer to Figure 14.83 in the textbook.

(a)

Calculate the parameters in frequency domain.

$$\begin{aligned} Z_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j\omega(9 \times 10^{-6})} \\ &= \frac{-j10^6}{9\omega} \Omega \end{aligned}$$

$$\begin{aligned} Z_L &= j\omega L \\ &= j\omega(20 \times 10^{-3}) \\ &= j0.02\omega \Omega \end{aligned}$$

$$Z_R = R$$

Step 2 of 5

Draw the circuit in frequency domain.

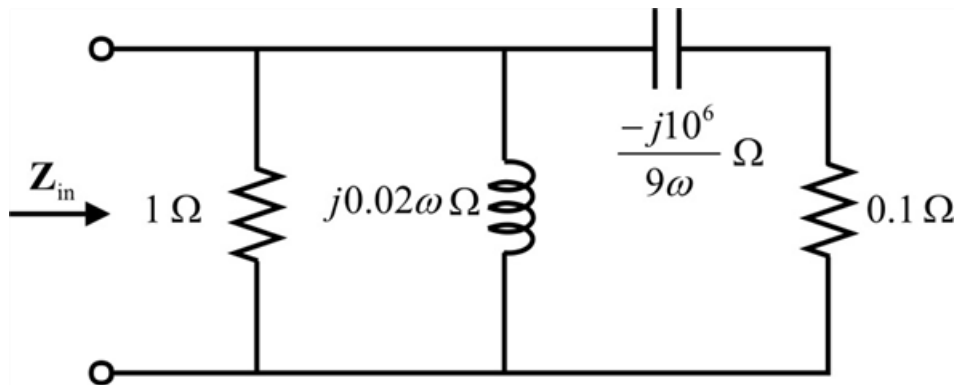


Figure 1

Step 3 of 5

Calculate the expression for $\mathbf{Z}_{in}$.

$$\begin{aligned}
 \mathbf{Z}_{in} &= (1) \parallel (j0.02\omega) \parallel \left(0.1 + \frac{-j10^6}{9\omega} \right) \\
 &= \left[\frac{(1) \times (j0.02\omega)}{1 + j0.02\omega} \right] \parallel \left(\frac{0.9\omega - j10^6}{9\omega} \right) \\
 &= \left(\frac{j0.02\omega}{1 + j0.02\omega} \right) \parallel \left(\frac{0.9\omega - j10^6}{9\omega} \right) \\
 &= \frac{\left(\frac{j0.02\omega}{1 + j0.02\omega} \right) \left(\frac{0.9\omega - j10^6}{9\omega} \right)}{\left(\frac{j0.02\omega}{1 + j0.02\omega} \right) + \left(\frac{0.9\omega - j10^6}{9\omega} \right)} \\
 &= \frac{(j0.02\omega)(0.9\omega - j10^6)}{j0.18\omega^2 + (0.9\omega - j10^6)(1 + j0.02\omega)} \\
 &= \frac{(j0.018\omega^2 + 20000\omega)}{j0.18\omega^2 + 0.9\omega + j0.018\omega^2 - j10^6 + 20000\omega} \\
 &= \frac{(j0.018\omega^2 + 20000\omega)}{20000.9\omega + j(0.198\omega^2 - 10^6)} \\
 &= \frac{(j0.018\omega^2 + 20000\omega) [20000.9\omega - j(0.198\omega^2 - 10^6)]}{(20000.9\omega)^2 + (0.198\omega^2 - 10^6)^2} \\
 &= \frac{j360.0162\omega^3 + 0.018\omega^2(0.198\omega^2 - 10^6) + 4 \times 10^8 \omega^2 - j20000\omega(0.198\omega^2 - 10^6)}{(20000.9\omega)^2 + (0.36\omega^2 - 10^6)^2} \\
 &= \frac{[0.018\omega^2(0.198\omega^2 - 10^6) + 4 \times 10^8 \omega^2]}{(20000.9\omega)^2 + (0.36\omega^2 - 10^6)^2} + j \frac{[360.0162\omega^3 - 20000\omega(0.198\omega^2 - 10^6)]}{(20000.9\omega)^2 + (0.36\omega^2 - 10^6)^2}
 \end{aligned}$$

Step 4 of 5

At resonant frequency, the imaginary term is equal to zero.

$$\frac{[360.0162\omega^3 - 20000\omega(0.198\omega^2 - 10^6)]}{(20000.9\omega)^2 + (0.36\omega^2 - 10^6)^2} = 0$$

$$360.0162\omega^3 - 20000\omega(0.198\omega^2 - 10^6) = 0$$

$$20000\omega(0.198\omega^2 - 10^6) = 360.0162\omega^3$$

$$0.198\omega^2 - 10^6 = \frac{360.0162\omega^2}{20000}$$

$$0.198\omega^2 - 10^6 = 0.018\omega^2$$

$$0.18\omega_0^2 - 10^6 = 0$$

$$\omega_0 = \sqrt{\frac{10^6}{0.18}}$$

$$\omega_0 = 2.357 \text{ kHz}$$

Thus, the value of resonant frequency is 2.357 kHz.

Step 5 of 5

At resonant frequency, the imaginary part of the impedance is zero. So, consider only the real part for impedance.

Calculate the impedance at resonant frequency, $\mathbf{Z_{in}}(\omega_0)$.

$$\mathbf{Z_{in}}(\omega_0) = \frac{0.018\omega^2(0.198\omega^2 - 10^6) + 4 \times 10^8 \omega^2}{(20000.9\omega)^2 + (0.36\omega^2 - 10^6)^2}$$

Substitute 2.357 kHz for ω_0 .

$$\begin{aligned}\mathbf{Z_{in}}(\omega_0) &= \frac{0.018(2.357 \times 10^3)^2 [0.198(2.357 \times 10^3)^2 - 10^6] + 4 \times 10^8 (2.357 \times 10^3)^2}{[20000.9(2.357 \times 10^3)]^2 + [0.36(2.357 \times 10^3)^2 - 10^6]^2} \\ &= \frac{2.222 \times 10^{15}}{2.223 \times 10^{15}} \\ &\approx 1 \, \Omega\end{aligned}$$

Thus, the value of $\mathbf{Z_{in}}(\omega_0)$ is $\boxed{1 \, \Omega}$.